

Ex. 3.14 | Ch. 3 | P. 93 | Ch/oe | 2025.7.24

证明: (a) $[\hat{A} + \hat{B}, \hat{C}] = (\hat{A} + \hat{B})\hat{C} - \hat{C}(\hat{A} + \hat{B}) = \hat{A}\hat{C} + \hat{B}\hat{C} - \hat{C}\hat{A} - \hat{C}\hat{B}$
 $= (\hat{A}\hat{C} - \hat{C}\hat{A}) + (\hat{B}\hat{C} - \hat{C}\hat{B}) = [\hat{A}, \hat{C}] + [\hat{B}, \hat{C}]$ 得证

左: $[\hat{A}\hat{B}, \hat{C}] = (\hat{A}\hat{B})\hat{C} - \hat{C}(\hat{A}\hat{B}) = \hat{A}\hat{B}\hat{C} - \hat{C}\hat{A}\hat{B}$

右: $\hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B} = \hat{A}(\hat{B}\hat{C} - \hat{C}\hat{B}) + (\hat{A}\hat{C} - \hat{C}\hat{A})\hat{B}$
 $= \hat{A}\hat{B}\hat{C} - \hat{A}\hat{C}\hat{B} + \hat{A}\hat{C}\hat{B} - \hat{C}\hat{A}\hat{B} = \hat{A}\hat{B}\hat{C} - \hat{C}\hat{A}\hat{B} = \text{左}, \text{得证}$

故: $[\hat{A} + \hat{B}, \hat{C}] = [\hat{A}, \hat{C}] + [\hat{B}, \hat{C}], [\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}$

(b) 证明 $[x^n, \hat{p}] = i\hbar n x^{n-1}$, 结合 $[x, \hat{p}] = i\hbar$, 使用归纳法

$n=1$ 时, $[x, \hat{p}] = i\hbar n x^{n-1} = i\hbar$ 成立

假设对 $n=k$ 有 $[x^k, \hat{p}] = i\hbar k x^{k-1}$ 成立

对 $n=k+1$, $[x^{k+1}, \hat{p}] = [x \cdot x^k, \hat{p}] = x[x^k, \hat{p}] + [x, \hat{p}]x^k$
 $= x \cdot (i\hbar k x^{k-1}) + (i\hbar)x^k = i\hbar(k+1)x^k$ 得证

(c) 证明 $[f(x), \hat{p}] = i\hbar \frac{df}{dx}$ ($f(x)$ 可泰勒展开)

$$[f(x), \hat{p}] = \left[\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n, \hat{p} \right] = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} [x^n, \hat{p}] = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} i\hbar n x^{n-1}$$

$$= i\hbar \sum_{n=1}^{\infty} \frac{f^{(n)}(0)}{(n-1)!} x^{n-1} = i\hbar \sum_{m=0}^{\infty} \frac{f^{(m+1)}(0)}{m!} x^m \quad (\triangleq m = n-1)$$

$$= i\hbar \frac{df}{dx} \quad \text{得证}$$

(d) 证明 $[\hat{H}, \hat{a}_{\pm}] = \pm \hbar \omega \hat{a}_{\pm}$

$\hat{H} = \hbar \omega \hat{a}_- \hat{a}_+ - \frac{1}{2} \hbar \omega$, $[\hat{a}_-, \hat{a}_+] = 1$, $[\hat{a}_+, \hat{a}_-] = -1$

$$[\hat{H}, \hat{a}_+] = [\hbar \omega \hat{a}_- \hat{a}_+ - \frac{1}{2} \hbar \omega, \hat{a}_+] = \hbar \omega [\hat{a}_- \hat{a}_+, \hat{a}_+] + [-\frac{1}{2} \hbar \omega, \hat{a}_+]$$

$$= \hbar \omega (\hat{a}_- [\hat{a}_+, \hat{a}_+] + [\hat{a}_-, \hat{a}_+] \hat{a}_+) + 0$$

$$= \hbar \omega (\hat{a}_- \cdot 0 + 1 \cdot \hat{a}_+) = \hbar \omega \hat{a}_+$$

$$[\hat{H}, \hat{a}_-] = [\hbar \omega \hat{a}_- \hat{a}_+ - \frac{1}{2} \hbar \omega, \hat{a}_-] = \hbar \omega [\hat{a}_- \hat{a}_+, \hat{a}_-] + [-\frac{1}{2} \hbar \omega, \hat{a}_-]$$

$$= \hbar \omega (\hat{a}_- [\hat{a}_+, \hat{a}_-] + [\hat{a}_-, \hat{a}_-] \hat{a}_+) + 0$$

$$= \hbar \omega (\hat{a}_- \cdot (-1) + 0 \cdot \hat{a}_+) = -\hbar \omega \hat{a}_-$$

故 $[\hat{H}, \hat{a}_{\pm}] = \pm \hbar \omega \hat{a}_{\pm}$ 得证